Arif Khan

LLM Engineer | Agentic AI Engineer | AI Engineer

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PROFESSIONAL SUMMARY

Agentic AI Engineer with 3+ years of experience building production multi-agent systems, LLM pipelines, and RAG architectures using LangChain, LangGraph, and LangSmith. Delivered a university production chatbot featuring Graph RAG, Hybrid Search, a 6-pipeline Supervisor workflow, and RAGAS evaluation serving 10 departments. Led a team of 3 to achieve Whole Tumor Dice exceeding 89% in a deep learning medical AI segmentation system at the University of Loralai. Capabilities span prompt engineering, fine-tuning (LoRA/QLoRA), vector databases, Docker containerization, and FastAPI deployment.

TECHNICAL SKILLS

LLM & Agentic AI:	LangChain, LangGraph, LangSmith, Multi-Agent Systems, Tool Calling, Prompt Engineering, Agentic Workflows, OpenAI API, Claude API, Hugging Face Transformers
RAG & Vector Databases:	RAG, Graph RAG, Hybrid Search (BM25 + Vector), ChromaDB, Redis, FAISS, Embeddings, Parent-Child Retrieval, Cross-Encoder Re-Ranking, Semantic Search
Evaluation & Observability:	RAGAS (Faithfulness, Answer Relevancy, Context Recall), LangSmith Tracing, Model Evaluation, Hallucination Detection
Deep Learning & ML:	PyTorch, TensorFlow, MONAI, Scikit-learn, XGBoost, CNNs, Computer Vision, Image Segmentation, Transfer Learning, Feature Engineering, NLP
MLOps, Backend & DevOps:	Docker, Containerization, GitHub Actions, FastAPI, Git, Streamlit, Web Scraping, Python, SQL
Data Analysis & Visualization:	NumPy, Pandas, Matplotlib, Seaborn, Plotly, SHAP Explainability, Data Pipelines

WORK EXPERIENCE

AI Engineer | University of Loralai February 2025 – October 2026

- Architected a production multi-agent LLM chatbot using LangChain and FastAPI, implementing a Supervisor pattern across 6 specialized pipelines (admissions, scheduling, academic support, FAQ) serving 10 university departments.
- Containerized the full system with Docker, ensuring consistent and reproducible deployments across development and production environments; maintained full infrastructure documentation and coordinated all release cycles.
- Implemented Graph RAG with BM25 hybrid search and Cross-Encoder re-ranking on ChromaDB and Redis, indexing 7,021documents across 500+ pages, significantly boosting retrieval precision and answer accuracy across all 10 departments.
- Elevated RAGAS and DeepEval evaluation scores through iterative prompt engineering and retrieval tuning: Faithfulness 0.75→0.85, Answer Relevancy 0.77→0.87, and Context Recall 0.60→0.85; monitored all LLM calls via LangSmith tracing to detect and reduce hallucinations.

- Coordinated with faculty stakeholders across all delivery phases from requirements scoping to production launch; communicated milestones biweekly and maintained full technical documentation across an 8-month engagement.

PROJECTS

University GPA Calculation — Agentic Workflow
Python, LangChain, LangGraph, SQLAlchemy, PostgreSQL, Streamlit — September 2025 – January 2026

- Built an autonomous LangGraph pipeline for GPA calculation, transcript parsing, and academic performance analysis; validated all calculation logic against official HEC Pakistan grading rubrics with 100% formula compliance.
- Delivered a Streamlit conversational interface enabling students to query academic standing, semester performance, and degreeprogress in natural language without manual lookup.
- Implemented role-based access control (Student, Coordinator, Controller of Examinations) with QR-verified PDF transcriptgeneration and full audit logging.

NeuroAI — Brain Tumor Segmentation System (Team Lead)
PyTorch, TensorFlow, MONAI, XGBoost, SHAP, FastAPI, Docker — January 2025 – July 2026

- Led a team of 3 to train a 3-model deep learning ensemble (DynUNet, SwinUNETR, SegResNet) on BraTS 2020 (369 cases); achieved Whole Tumor Dice >89%, Tumor Core Dice >85%, and Enhancing Tumor Dice >77%.
- Built an interactive 3D MRI viewer, automated PDF clinical reports, multi-year scan history comparison, and an XGBoost gradeclassifier with SHAP explainability.
- Extracted 30 quantitative radiomics features per scan and deployed the full clinical AI pipeline to a Streamlit server with Dockercontainerization, enabling end-to-end inference without manual intervention.

Student Performance Predictor — 98% R^2 Score
Python, Scikit-learn, XGBoost, Feature Engineering, Streamlit — January 2024 – October 2024

- Engineered a machine learning regression system to predict student academic performance, achieving 98% R^2 score; evaluatedRandom Forest, XGBoost, and Linear Regression with cross-validation and hyperparameter tuning.
- Conducted thorough data preprocessing, feature engineering, and exploratory data analysis to identify the optimal modelconfiguration.
- Shipped an interactive Streamlit dashboard enabling educators to receive instant at-risk predictions with SHAP-based featureimportance visualizations to support early intervention.

Laptop Price Predictor
Python, Scikit-learn, Random Forest, Feature Engineering, Streamlit — July 2023 – December 2023

- Extracted 8 engineered features from laptop records and benchmarked 5 regression models; Random Forest achieved R^2=0.88with the lowest mean absolute error across all candidates.
- Performed end-to-end feature engineering including brand encoding, CPU tier classification, and storage type parsing, producinga generalized model accurate across 20+ laptop manufacturers.
- Deployed as a Streamlit web application with 12 configurable hardware inputs, achieving under 500ms response time forconsistent real-time price estimation.

EDUCATION

BS Computer Science — University of Loralai — August 2022 – May 2026

CGPA: 3.73 / 4.0

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Structures and Algorithms, Database Systems, Software Engineering